

Inés Alejandro Cruz Guerrero

Postdoctoral Fellow

Current Position

Postdoctoral Fellow

Department of Neurological Surgery, Biological Sciences Division,
University of Chicago, Chicago, IL, USA.

Responsibilities

- Development of deep learning and computer vision methods applied to medical image analysis and photogrammetry studies.
- Design and implementation of image processing and analysis algorithms for feature extraction and three-dimensional modeling.
- Preparation and submission of scientific publications to peer-reviewed journals and specialized conferences.
- Mentoring and supervision of graduate and undergraduate students in research projects.
- Coordination with clinical and technical collaborators for the validation and application of developed methods.
- Contribution to the planning and management of research laboratory activities.

Contact Information

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Google Scholar: <https://scholar.google.com/citations?user=6WnxDuUAAAAJ&hl=en>

Education

2019–2023 **DOCTORATE**, *School of Science, Autonomous University of San Luis Potosí (UASLP)*, S.L.P., Mexico

PhD in Engineering Sciences with Emphasis on Biomedical Engineering

Dissertation: *Advanced Blind Decomposition Methods for End-Members and Abundance Estimation in Medical Imaging*.

Graduated with Honors.

2017–2019 **MASTER'S DEGREE**, *School of Science, Autonomous University of San Luis Potosí (UASLP)*, S.L.P., Mexico

Master's in Electronics Engineering with Emphasis on Bioelectronics

Thesis: *Quantification of Subcutaneous and Visceral Fat in Medical Images*.

2012–2017 **BACHELOR'S DEGREE**, *School of Science, Autonomous University of San Luis Potosí (UASLP)*, S.L.P., Mexico

Bachelor's in Biomedical Engineering

Thesis: *Holographic Visualization System for Medical Images*.

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1/9

Research Areas

Optimization Algorithms, Digital Signal and Medical Image Processing, Deep Learning Algorithms, Computer Vision on Meshes and Complex Geometries, 3D Image Reconstruction and Segmentation, Generative Models and Graph Neural Networks

Professional and Academic Experience

- Mar 2026 – **Postdoctoral Fellow**, *Department of Neurological Surgery, Biological Sciences Division*, University of Chicago, Chicago, IL, USA
Present
- Oct 2023 – **Postdoctoral Researcher**, *Department of Biostatistics and Informatics, Colorado School of Public Health*, University of Colorado Anschutz Medical Campus, Aurora, CO, USA
Mar 2026
- Mar 2022 – **Adjunct Faculty**, *Course: Physics*, Instituto Latinoamericano de San Luis, S.L.P., Mexico
Jun 2023
- Sep 2021 – **Research Internship**, *Institute for Applied Microelectronics (IUMA)*, University of Las Palmas de Gran Canaria, Spain
Nov 2021
- Aug 2020 – **Adjunct Faculty**, *Course: Electricity and Magnetism, Bachelor's in Biomedical Engineering*, School of Science, UASLP, Mexico
Dec 2020
- Aug 2018 – **Social Service**, *Laboratory of Basic Biomedical Engineering*, School of Science, UASLP, Mexico
Feb 2019
- Oct 2016 – **Professional Internship**, *Laboratory of Basic Biomedical Engineering*, School of Science, UASLP, Mexico
Feb 2017
- Mar 2016 – **Social Service**, *Clinical Engineering Department*, TECNOS Clinical Hospital Engineering, San Luis Potosí, Mexico
Sep 2016

Awards and Honors

- 2024 **Honorable Mention**, *Outstanding Reviewer, 27th International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI 2024)*, Marrakesh, Morocco.
- 2023 **Honor**, *PhD in Engineering Sciences graduated with honors*, Autonomous University of San Luis Potosí.
- 2022 **First Prize**, “Dr. Fernando Prieto Hernández” graduate student competition, *National Congress of Biomedical Engineering 2022*, Puerto Vallarta, Jalisco.
- 2021 **Mobility Scholarship**, CUMex – AUIP 2021 Program for international research internship.
- 2020 **Second Prize**, “Ing. Teófila Cadena Alfaro” graduate student competition, *National Congress of Biomedical Engineering 2020* (virtual).
- 2017 **First Prize**, “InnovaHealth Challenge” undergraduate student competition, *National Congress of Biomedical Engineering 2017*, Monterrey, Nuevo León.

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2/9

Research Projects

- “Data-driven Quantification and Prediction of Pre-Surgical Local Head Volume Distributions and Post-Surgical Development in Patients with Craniosynostosis”, National Institutes of Health (NIH), Grant No. R01DE032681. Role: Assistant Researcher.
- “Blind Estimation Under Spatial Coherence and Mixed Noise” (Estimación Ciega Bajo Coherencia Espacial y Ruido Mezclado), CONACYT Basic Science Grant 2015, No. 254637. Role: Assistant Researcher.

Additional Training

- 2023 **Certificate**, *Group 1 Biomedical Investigators*, CITI Program
- 2023 **Certificate**, *Conflict of Interest*, CITI Program
- 2023 **Certificate**, *Health Information Privacy and Security (HIPS) for Clinical Investigators*, CITI Program
- 2023 **Certificate**, *Biomedical Responsible Conduct of Research*, CITI Program
- 2023 **Certificate**, *ICH/GCP – Meets NIH Requirements*, CITI Program
- 2022 **Workshop**, *Data Science Open*, BEDU, Duration: 1 month
- 2021 **Certificate**, *Microsoft Azure Fundamentals (AZ-900)*, Microsoft, Duration: 3 months
- 2020 **Workshop**, *Spectral Tomography*, CVIB 2020, Duration: 8 h
- 2017 **Workshop**, *Digital Processing with LabVIEW*, ENIBET 2017, Duration: 4 h
- 2017 **Workshop**, *New Technologies: Electrical Impedance Tomography*, CNIB 2017, Duration: 4 h
- 2017 **Workshop**, *Remote Service for Imaging Systems*, CNIB 2017, Duration: 4 h
- 2017 **Course**, *CCNA Routing & Switching: Introduction to Networks*, Cisco Networking Academy, Duration: 6 months
- 2015 **Workshop**, *Features of Anesthesia Systems for Hospital Savings and Minimal Contamination*, CNIB 2015, Duration: 4 h
- 2014 **Workshop**, *Introduction to Neuro-navigation by Magnetic Resonance*, CNIB 2014, Duration: 4 h
- 2013 **Workshop**, *Introduction to Biomaterials Science and Engineering*, CNIB 2013, Duration: 8 h

Volunteer Services

- 2025 **Peer Reviewer**, 28th International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI 2025), Daejeon, South Korea.
- 2024 **Peer Reviewer**, 27th International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI 2024), Marrakesh, Morocco.
- 2024 – **Peer Reviewer**, Journal of the Franklin Institute
Present
- 2023 – **Peer Reviewer**, IEEE Access
Present

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2023 – **Peer Reviewer**, IEEE Latin America Transactions
Present

2022 – **Special Session Committee Member**, “*Hyperspectral Imaging Applications, Algorithms and Architectures*”, Euromicro Conference on Digital System Design (DSD)
Present

PhD Dissertation Committee

2024 **Abián Hernández Guedes**, *PhD in Telecommunication Technologies and Computer Engineering*, Institute for Applied Microelectronics (IUMA), University of Las Palmas de Gran Canaria (ULPGC)

2024 **Raquel León Martín**, *PhD in Telecommunication Technologies and Computer Engineering*, Institute for Applied Microelectronics (IUMA), University of Las Palmas de Gran Canaria (ULPGC)

Journal Publications

- **Cruz-Guerrero, I. A.**, Nagel, J., & Porras, A. R. (2026). “Unsupervised Adaptive Sampling Graph Autoencoder for 3D Surface Encoding and Mesh Representation Transfer”. *Medical Image Analysis*.
- Palmer, S. K., Elkhill, C., **Cruz-Guerrero, I. A.**, Gomez, D. A., Linguraru, M. G., Wilkinson, C., Khechoyan, D., Nguyen, P., French, B., & Porras, A. R. (2026). “Data-Driven Personalized Surgical Treatment Selection for Metopic and Sagittal Craniosynostosis”. *Journal of Neurosurgery: Pediatrics*, 1–10.
- Liu, J., Xing, F., Elkhill, C., Linguraru, M. G., Miles, R. C., **Cruz-Guerrero, I. A.**, & Porras, A. R. (2025). “Population-Driven Synthesis of Personalized Cranial Development from Cross-Sectional Pediatric CT Images”. *IEEE Transactions on Biomedical Engineering*, 72(9), 2732–2741.
- Palmer, S. K., Elkhill, C., **Cruz-Guerrero, I. A.**, Gomez, D. A., Wilkinson, C., Khechoyan, D., French, B., Porras, A. R., & Nguyen, P. (2025). “Early Outcomes of Endoscopic Spring-Assisted Repair of Metopic Craniosynostosis”. *Journal of Craniofacial Surgery*, 36(5), 1480.
- Mendoza-Chavarría, J. N., **Cruz-Guerrero, I. A.**, Gutiérrez-Navarro, O., León, R., Ortega, S., Fabelo, H., Callicó, G. M., & Campos-Delgado, D. U. (2024). “Blind Non-Linear Spectral Unmixing with Spatial Coherence for Hyper and Multispectral Images”. *Journal of the Franklin Institute*, 361(18), 107282.
- Görg, C., Elkhill, C., Chaij, J., Royalty, K., Nguyen, P. N., French, B., **Cruz-Guerrero, I. A.**, & Porras, A. R. (2024). “SHAPE: A Visual Computing Pipeline for Interactive Landmarking of 3D Photograms and Patient Reporting for Assessing Craniosynostosis”. *Computers & Graphics*, 125, 104056.
- **Cruz-Guerrero, I. A.**, Campos-Delgado, D. U., Mejía-Rodríguez, A. R., León, R., Ortega, S., Fabelo, H., & Callicó, G. M. (2024). “Hybrid Brain Tumor Classification of Histopathology Hyperspectral Images by Linear Unmixing and an Ensemble of Deep Neural Networks”. *Healthcare Technology Letters*, 11(4), 240–251.
- **Cruz-Guerrero, I. A.**, Mejía-Rodríguez, A. R., Ortega, S., Fabelo, H., Callicó, G. M., Jo, J. A., & Campos-Delgado, D. U. (2023). “Multi and Hyperspectral Image Unmixing with Spatial Coherence by Extended Blind End-Member and Abundance Extraction”. *Journal of the Franklin Institute*, 360(15), 11165–11196.
- León, R., Fabelo, H., Ortega, S., **Cruz-Guerrero, I. A.**, Campos-Delgado, D. U., Szolna, A., & Callicó, G. M. (2023). “Hyperspectral Imaging Benchmark Based on Machine Learning for Intraoperative Brain Tumour Detection”. *npj Precision Oncology*, 7(1), 119.
- Campos-Delgado, D. U., **Cruz-Guerrero, I. A.**, Mendoza-Chavarría, J. N., Mejía-Rodríguez, A. R., Ortega, S., Fabelo, H., & Callicó, G. M. (2022). “Nonlinear Extended Blind End-Member and Abundance Extraction for Hyperspectral Images”. *Signal Processing*, 201, 108718.
- **Cruz-Guerrero, I. A.**, León, R., Campos-Delgado, D. U., Ortega, S., Fabelo, H., & Callicó, G. M. (2020). “Classification of Hyperspectral In Vivo Brain Tissue Based on Linear Unmixing”. *Applied Sciences*, 10(16), 5686.

Book Chapters

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5/9

- **Cruz-Guerrero, I. A.**, León, R., Mejía-Rodríguez, A. R., Campos-Delgado, D. U., Fabelo, H., Ortega, S., & Callicó, G. M. (2023). “Hyperspectral Imaging for Cancer Applications”. In *Diagnosis and Treatment of Cancer Using Thermal Therapies*, pp. 81–101. CRC Press.
- **Cruz-Guerrero, I. A.**, Campos-Delgado, D. U., Mejía-Rodríguez, A. R., Fabelo, H., Ortega, S., & Callicó, G. M. (2022). “A Hybrid Approach to the Hyperspectral Classification of In Vivo Brain Tissue: Linear Unmixing with Spatial Coherence and Machine Learning”. In *Artificial Intelligence in Cancer Diagnosis and Prognosis: Brain and Prostate Cancer*, Vol. 3, Ch. 9, pp. 9–1–9–22. IOP Publishing.

International Conference Publications

- Elkhill, C., **Cruz-Guerrero, I. A.**, Khechoyan, D., Nguyen, P. N., French, B., & Porras, A. R. (2026). “A Temporally Aware Transformer Model for Predicting Personalized Head Growth and Surgical Outcomes in Craniosynostosis”. In *2026 IEEE 23rd International Symposium on Biomedical Imaging (ISBI)*, pp. 1–4. IEEE.
- Elkhill, C., **Cruz-Guerrero, I. A.**, Linguraru, M. G., French, B., & Porras, A. R. (2025). “A Generative Model with Domain Adversarial Training and Independent Conditional Discriminators to Predict the Development of Craniosynostosis”. In *2025 21st International Symposium on Medical Information Processing and Analysis (SIPAIM)*, pp. 1–4. IEEE.
- Elkhill, C., **Cruz-Guerrero, I. A.**, Liu, J., Linguraru, M. G., Alexander, A., French, B., & Porras, A. R. (2024). “Personalized Phenotype Encoding and Prediction of Pathological Head Development from Cross-Sectional Images”. In *2024 20th International Symposium on Medical Information Processing and Analysis (SIPAIM)*, pp. 1–4. IEEE.
- **Cruz-Guerrero, I. A.**, Elkhill, C., Liu, J., Nguyen, P. N., French, B., & Porras, A. R. (2024). “Mesh Registration via Geometric Feature Homogenization and Offset Cross-Attention: Application to 3D Photogrammetry”. In *International Workshop on Graphs in Biomedical Image Analysis (GRAIL 2024)*, pp. 96–105. Springer, Cham.
- Mendoza-Chavarría, J. N., **Cruz-Guerrero, I. A.**, Campos-Delgado, D. U., Fabelo, H., Ortega, S., & Callicó, G. M. (2023). “Ensemble of Artificial Intelligence Classifiers for In-Vivo Identification of Glioblastoma Tumours Using Hyperspectral Images”. In *2023 1st IEEE Engineering in Medicine and Biology Conference in Latin America (IEEE EMBC R9)*. IEEE.
- **Cruz-Guerrero, I. A.**, Mendoza-Chavarría, J. N., Campos-Delgado, D. U., Fabelo, H., Ortega, S., & Callicó, G. M. (2023). “Classification of Brain Tissues in Hyperspectral Images Using Vision Transformers”. In *2023 IEEE 20th International Symposium on Biomedical Imaging (ISBI)*. IEEE.
- **Cruz-Guerrero, I. A.**, León, R., Granados-Castro, L., Fabelo, H., Ortega, S., Campos-Delgado, D. U., & Callicó, G. M. (2022). “Reflectance Calibration with Normalization Correction in Hyperspectral Imaging”. In *2022 25th Euromicro Conference on Digital System Design (DSD)*. IEEE.
- Mendoza-Chavarría, J. N., Zavala-Sánchez, E. R., Granados-Castro, L., **Cruz-Guerrero, I. A.**, Fabelo, H., Ortega, S., Callicó, G. M., & Campos-Delgado, D. U. (2022). “Glioblastoma Classification in Hyperspectral Images by Nonlinear Unmixing”. In *2022 25th Euromicro Conference on Digital System Design (DSD)*. IEEE.
- Granados-Castro, L., Gutiérrez-Navarro, O., **Cruz-Guerrero, I. A.**, Mendoza-Chavarría, J. N., Zavala-Sánchez, E. R., & Campos-Delgado, D. U. (2022). “Estimation of Deoxygenated and Oxygenated Hemoglobin by Multispectral Blind Linear Unmixing”. In *2022 25th*

Euromicro Conference on Digital System Design (DSD). IEEE.

- **Cruz-Guerrero, I. A.**, Campos-Delgado, D. U., & Mejía-Rodríguez, A. R. (2021). “Extended Blind End-Member and Abundance Estimation with Spatial Total Variation for Hyperspectral Imaging”. In *2021 43rd Annual International Conference of the IEEE Engineering in Medicine & Biology Society (EMBC)*, pp. 1957–1960. IEEE.
- Alvarez-Gutiérrez, M. T., Mejía-Rodríguez, A. R., **Cruz-Guerrero, I. A.**, & Arce-Santana, E. R. (2019). “3D Kidney Reconstruction from 2D Ultrasound Images”. In *Latin American Conference on Biomedical Engineering*, pp. 393–400. Springer, Cham.

■ National Conference Publications

- Martínez-Arboleyda, D. N., **Cruz-Guerrero, I. A.**, & Campos-Delgado, D. U. (2023). “Tumor Tissue Classification in Hyperspectral Histopathology Images Through Individual and Ensemble of Machine Learning Algorithms”. In *XLVI Mexican Conference on Biomedical Engineering (CNIB 2023)*, pp. 100–111. Springer, Cham.
- **Cruz-Guerrero, I. A.**, Mendoza-Chavarría, J. N., & Campos-Delgado, D. U. (2022). “Glioblastoma Classification in Hyperspectral Images by Reflectance Calibration with Normalization Correction and Nonlinear Unmixing”. In *XLV Mexican Conference on Biomedical Engineering (CNIB 2022)*, Vol. 86. Springer, Cham.
- **Cruz-Guerrero, I. A.**, Campos-Delgado, D. U., Mejía-Rodríguez, A. R., Ortega, S., Fabelo, H., & Marrero Callicó, G. M. (2021). “Hybrid Approach to Histological Specimen Classification for Brain Tumor Detection: Linear Unmixing and Neural Networks”. In *XLIV Mexican Conference on Biomedical Engineering (CNIB 2021)*, 8(1), pp. 97–100.
- Mendoza-Chavarría, J. N., **Cruz-Guerrero, I. A.**, Mejía-Rodríguez, A. R., & Campos-Delgado, D. U. (2021). “Blind Unmixing Algorithm Based on the Multi-Linear Mixture Model”. In *XLIV Mexican Conference on Biomedical Engineering (CNIB 2021)*, 8(1), pp. 106–109.
- **Cruz-Guerrero, I. A.**, Campos-Delgado, D. U., & Mejía-Rodríguez, A. R. (2020). “Blind Estimation of Total Variance Endmembers and Abundances for Hyperspectral Imaging for Clinical Use”. In *XLIII Mexican Conference on Biomedical Engineering (CNIB 2020)*, 7(1), pp. 548–555.

■ Oral Presentations

- **Oral presentation**, “Mesh Registration via Geometric Feature Homogenization and Offset Cross-Attention: Application to 3D Photogrammetry”, *International Workshop on Graphs in Biomedical Image Analysis (GRAIL)*, MICCAI, 2024.
- **Poster presentation**, “Classification of Brain Tissues in Hyperspectral Images Using Vision Transformers”, *IEEE International Symposium on Biomedical Imaging (ISBI)*, 2023.
- **Oral presentation**, “Reflectance Calibration with Normalization Correction in Hyperspectral Imaging”, *Euromicro Conference on Digital System Design (DSD)*, 2022.
- **Oral presentation**, “Glioblastoma Classification in Hyperspectral Images by Reflectance Calibration with Normalization Correction and Nonlinear Unmixing”, *National Congress of Biomedical Engineering (CNIB)*, 2022.
- **Oral presentation**, “Extended Blind End-Member and Abundance Estimation with Spatial Total Variation for Hyperspectral Imaging”, *IEEE Engineering in Medicine and Biology Society Annual Conference (EMBC)*, 2021.
- **Oral presentation**, “Blind Estimation of Profiles and Abundances with Total Variation for

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7/9

Hyperspectral Images for Clinical Use”, *Seminar on Biomedical, Electronic and Telecommunications Engineering*, School of Science, UASLP, 2020.

- **Poster presentation**, “Quantification of Subcutaneous and Visceral Fat in Medical Images”, *2nd Encounter of Biomedical, Electronic and Telecommunications Engineering*, School of Science, UASLP, 2018.
- **Oral presentation**, “Wireless Device for Physiological Signal Acquisition in Clinical Applications”, *4th Encounter of Young Researchers in San Luis Potosí State*, San Luis Potosí, 2016.

Technical Skills

- Programming Languages: Python, C++, C, C#, VHDL, MATLAB
- AI/ML Frameworks: PyTorch, TensorFlow, Scikit-learn, NumPy, SciPy
- Software Libraries and Tools: OpenCV, VTK, MIPAV, LabVIEW, Simulink, BioPac, Multi-sim, PCBWizard, Easy-PC, SolidWorks
- Productivity Tools: Microsoft Office, LibreOffice, L^AT_EX, Mendeley, Zotero, OBS Studio
- Languages: Spanish (Native), English (B2)

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9/9